THE TANGENT SPACE OF THE CHANGE OF RESIDUE FIELD OF A PREDEFORMATION CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-defos.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to the associated category in sets with the chain of equalities identifying the tangent space of the changed category, the generalization to trivial extensions by finite dimensional vector spaces, and the application of uniqueness of a linear structure.

Editorial note. The label lemma-tangent-space-change-of-field names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source formal-defos.tex –label lemma-tangent-space-change-of-field –start 7598 –end 7632 –local-id 28029 returned RESULT=OK (L28029 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-formal-defos/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Elementary properties of the change of residue field construction. Same chapter, the lemma collecting the elementary properties of the change of residue field of a category cofibred in groupoids. Supplies the reduction to the associated category in sets and the identification of the deformation situations.
- Vector space structure on the tangent space and its description by the value at the trivial extension. Same chapter, the lemma providing the natural vector space structure on the tangent space and identifying its elements with the values of the functor on the trivial extension. Supplies the chain of equalities computing the tangent space of the changed category.
- Uniqueness of a linear structure compatible with a functor. Same chapter, the lemma that a functor to the category of sets admits at most one structure of a linear functor. Supplies the conclusion that the identifications are compatible with the l-vector space structures.

Source reference. Original reference 07WA: sources/stacks-formal-defos/pinned-original-source.tex, source lines 7539–7539. Prerequisite pointer invoked at proof line(s) 7611; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:formal-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 06I6: sources/stacks-formal-defos/pinned-original-source.tex, source lines 3369–3369. Prerequisite pointer invoked at proof line(s) 7628; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06IH: sources/stacks-formal-defos/pinned-original-source.tex, source lines 3669–3669. Prerequisite pointer invoked at proof line(s) 7615; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 07W8: sources/stacks-formal-defos/pinned-original-source.tex, source lines 7494–7494. Prerequisite pointer invoked at proof line(s) 7600; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With notation and assumptions as in Situation 07W8. Assume $\mathcal{F}$ is a predeformation category and $\overline{\mathcal{F}}$ satisfies (S2). Then there is a canonical l-vector space isomorphism*

$$T\mathcal{F} \otimes_k l \longrightarrow T\mathcal{F}_{l/k}$$

of tangent spaces.

Proof. By Lemma 07WA we may replace $\mathcal{F}$ by $\overline{\mathcal{F}}$. Moreover we see that $T\mathcal{F}$, resp. $T\mathcal{F}_{l/k}$ has a canonical k-vector space structure, resp. l-vector space structure, see Lemma 06IH. Then

$$T\mathcal{F}_{l/k} = \mathcal{F}_{l/k}(l[\epsilon]) = \mathcal{F}(k[l\epsilon]) = T\mathcal{F} \otimes_k l$$

the last equality by Lemma 06IH. More generally, given a finite dimensional l-vector space V we have

$$\mathcal{F}_{l/k}(l[V]) = \mathcal{F}(k[V_k]) = T\mathcal{F} \otimes_k V_k$$

where V_k denotes V seen as a k-vector space. We conclude that the functors $V \mapsto \mathcal{F}_{l/k}(l[V])$ and $V \mapsto T\mathcal{F} \otimes_k V_k$ are canonically identified as functors to the category of sets. By Lemma 06I6 we see there is at most one way to turn either functor into an l-linear functor. Hence the isomorphisms are compatible with the l-vector space structures and we win. □